

3.3 The characteristics of the tropical rainforest ecosystem

This chapter will explain:

- ★ the global distribution of tropical rainforests
- ★ the characteristics of the equatorial climate
- ★ the characteristics and structure of the tropical rainforest ecosystem.

The global distribution of tropical rainforests

Evergreen **tropical rainforests** are located in equatorial areas, largely between 10°N and 10°S ([Figure 3.10](#)). There are some areas of rainforest found outside these regions, but these tend to be more seasonal in nature. The main areas of rainforest include the Amazon rainforest in Brazil, the Congo rainforest in central Africa and the Indonesian–Malaysian rainforests of Southeast Asia. There are also many small fragments of rainforest, such as those on the island of Madagascar and in the Caribbean. Tropical rainforests everywhere are under increasing threat from human activities such as farming and logging. The result is that rainforests are disappearing and those that remain are not only smaller, but are broken up into fragments.

Interesting note

Tropical rainforests cover 6 per cent of the world's land surface but hold 50 per cent of the world's species. The Amazon rainforest alone is home to 10 per cent of the world's known species.



▲ **Figure 3.10** World distribution of tropical rainforests

The characteristics of the equatorial climate

The main characteristics of an equatorial climate include:

- » hot conditions – generally above 26°C – throughout the year, as the overhead sun is always high in the sky by the middle of the day
- » high levels of rainfall, often over 2000 mm ([Figure 3.11](#))
- » tropical rainforests characterised by low pressure – rising air that cools and produces clouds and rains
- » a lack of seasons – the temperatures are high throughout the year
- » the difference between daytime and night-time temperatures (known as the diurnal range) is higher than the seasonal differences in temperature
- » rainfall is mainly convectional – heating causes air to rise and then cool, and rain may fall on as many as 250 days each year
- » cloud cover varies – in the morning it may be limited, but by the afternoon towering cumulonimbus clouds mark the start of the **convectional rains**
- » the presence of clouds tends to reduce the amount of heat that is lost at night – hence the diurnal range is less than in hot desert areas
- » the **humidity** (moisture in the atmosphere) is high, and relative humidities of 100 per cent are often

reached in the late afternoon

- » wind speeds within the rainforest are reduced by the large numbers of trees present
- » with increasing latitude there is increased seasonality of the rainforest.



▲ **Figure 3.11** Convectional uplift causes cloud and rain on most days in a tropical rainforest, such as here in Sarawak

The data for Manaus in Brazil ([Table 3.3](#)) show that the warmest months are September and October, with a maximum temperature of 34°C. In contrast, all of the months from December to September share the minimum temperature of 24°C. Thus the annual temperature range is 10°C.

Rainfall in Manaus is high, nearly 2100 mm annually. There is a definite wet season between November and May, whereas the months of June to October are relatively dry.

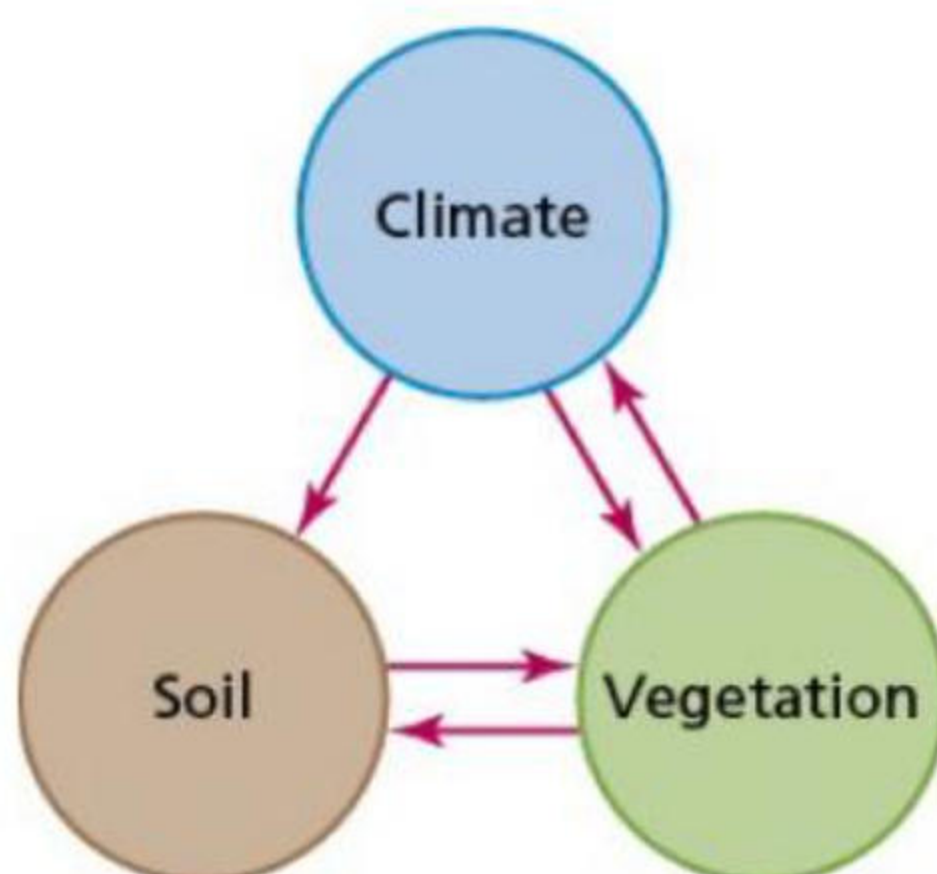
Activities

- 1 Calculate for Manaus:
 - a the mean average monthly temperature
 - b the annual precipitation.
 - 2 Compare the seasonal variations in rainfall and sunshine (hours) for Manaus.
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The characteristics and structure of the tropical rainforest ecosystem

The tropical rainforest ecosystem includes biotic factors (such as plants and animals) and abiotic factors (such as heat, rainfall, oxygen and minerals). These are closely interrelated ([Figure 3.12](#)).

The links between climate, soil and vegetation are very strong



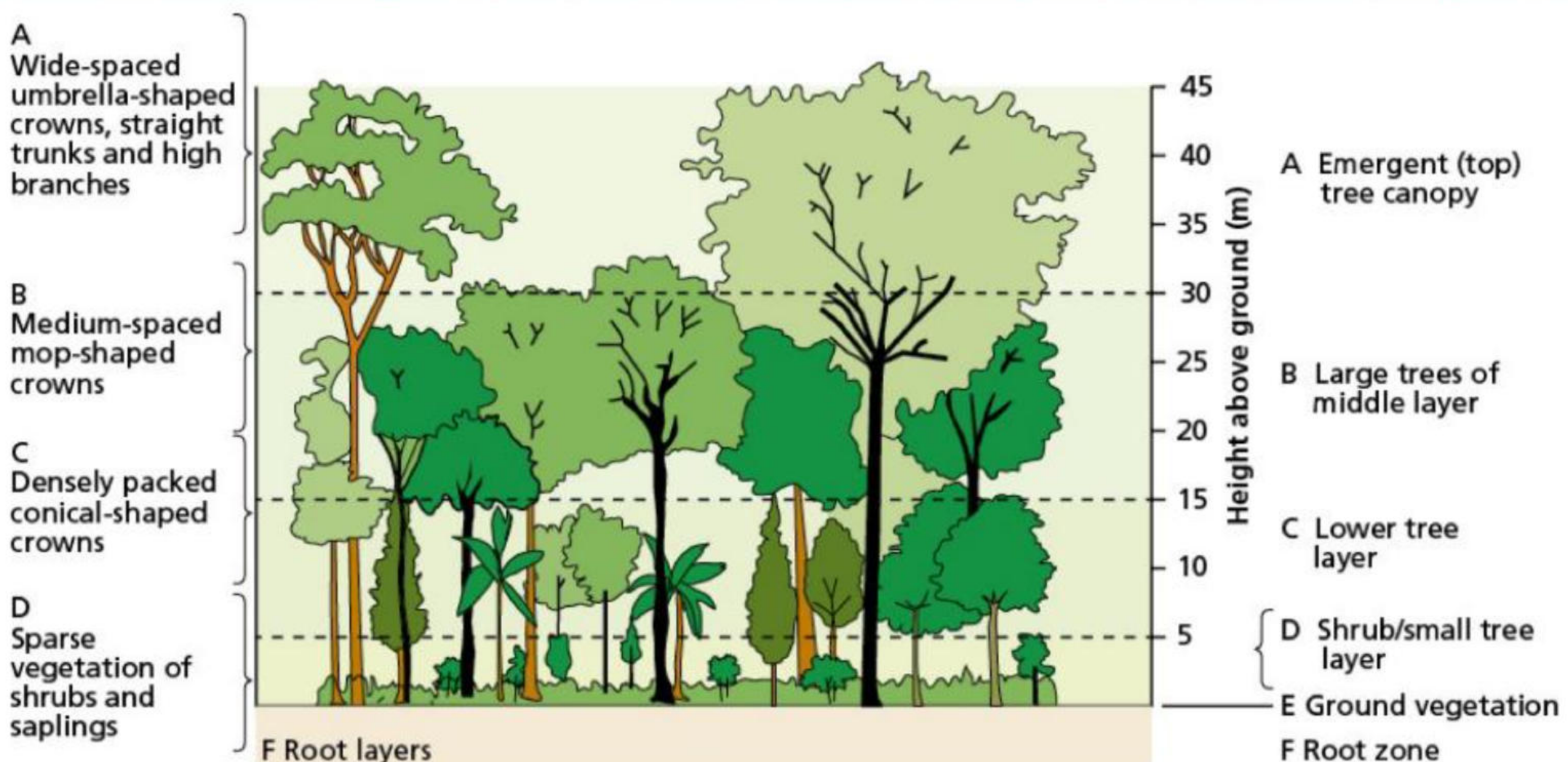
▲ **Figure 3.12** The links between climate, soil and vegetation in a tropical rainforest ecosystem

Vegetation is evergreen, enabling photosynthesis to take place all year round. This is possible due to the high temperatures and presence of water throughout the year. The vegetation is layered, and the shape of the crowns varies at each layer ([Figure 3.13](#)). Species at the top of the canopy receive most of the sunlight, whereas species that are located near the forest floor are adapted to darker conditions and generally have a darker pigment so as to photosynthesise at low light levels. There is a great variety in the number of species in a rainforest – this is known as **biodiversity**. A single hectare in a rainforest may contain as

many as 300 different species. Typical rainforest species include figs, teak, mahogany and yellow woods.

▼ **Table 3.3** Climate data for Manaus, Brazil

	J	F	M	A	M	J	J	A	S	O	N	D	Av/Total
Temperature:													
Daily max (°C)	31	31	31	31	31	31	32	33	34	34	33	32	32
Daily min (°C)	24	24	24	24	24	24	24	24	24	25	25	24	24
Average monthly (°C)	28	28	28	27	28	28	28	29	29	29	29	28	
Rainfall:													
Monthly total (mm)	278	278	300	287	193	99	61	41	62	112	165	220	
Sunshine:													
Sunshine (hours)	3.9	4	3.6	3.9	5.4	6.9	7.9	8.2	7.5	6.6	5.9	4.9	5.7



▲ **Figure 3.13** Vegetation structure of the tropical rainforest

Soils

Rainforests are the most productive land-based ecosystems. Ironically, the soils of tropical rainforests are quite infertile. This is because most of the nutrients in the rainforest are contained in the biomass (living matter). Rainforest soils are typically deep due to the large amount of weathering that has taken place, and they are often red in colour due to large amounts of iron being present in the soil. Nevertheless, there are some areas in which tropical soils may be more fertile: in floodplains and in volcanic areas the soils may be enriched by flooding or by the weathering of fertile lava flows.

How flora and fauna adapt to survive

Flora

Tropical vegetation in rainforests has many adaptations. Some trees have leaves with drip-tips ([Figure 3.14a](#)), which are designed to get rid of excess moisture. In contrast, other plants have saucer-shaped leaves in order to collect water. Pitcher plants have developed an unusual means of getting their nutrients. Rather than taking nutrients from the soil, they have become carnivorous and get their nutrients from insects and small frogs that are trapped inside the pitcher ([Figure 3.14b](#)). This is one way of coping with the very infertile soils of the rainforest. Other plants are very tall. To prevent being blown over by the wind, very large trees have developed buttress roots that project out from the main trunk above the ground, which gives the plant extra leverage in the wind.



▲ **Figure 3.14** Adaptations of rainforest plants: a drip-tip; b pitcher plant

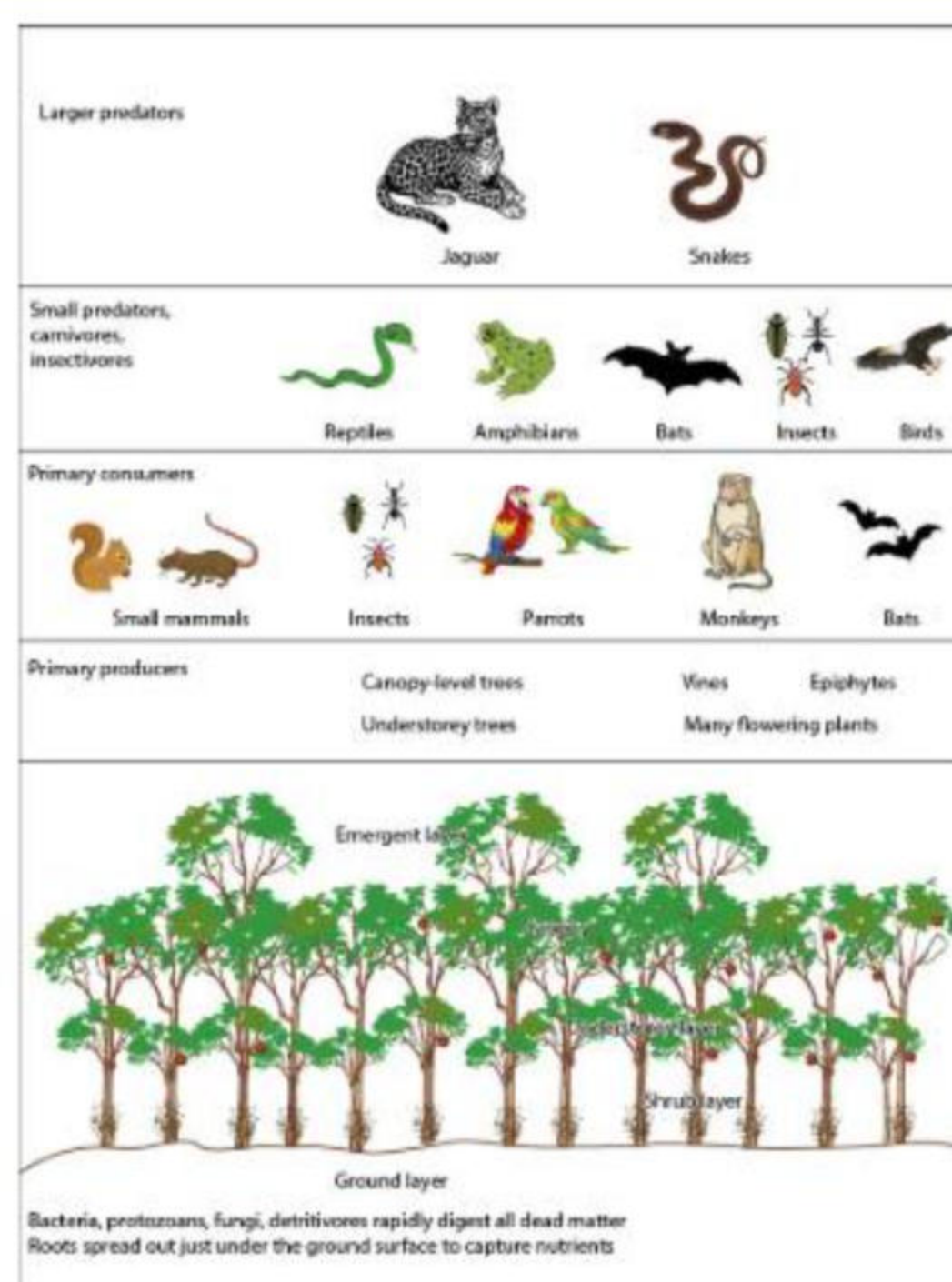
Fauna

Although rainforests cover less than 6 per cent of the Earth's surface, they account for over 50 per cent of all animal species on Earth. Tropical rainforests are very biodiverse ([Figure 3.15](#)). Many species, such as orangutans, are arboreal (they live entirely in the trees) and rarely come to the ground, as they would be easy prey for large carnivores there.

Top carnivores, such as jaguars, tigers and leopards, are highly camouflaged to aid in hunting. Many other species are highly camouflaged to avoid becoming prey, such as stick insects and the Indian oakleaf butterfly. Sloths are covered with a layer of green algae that camouflages their fur in their arboreal environment. Some animals have evolved to look larger or more scary than they really are: the larvae of the lobster moth look like scorpions, but they are defenceless. Many butterflies have designs that look like large eyes on their wings, in order to confuse potential predators. Some species, such as the monarch butterfly, are poisonous, which helps deter predators. It is more than likely that there are many rainforest species that have not yet been discovered – according to the World Wide Fund for Nature, new species are being discovered in the Amazon 'every other day'. Between 2016 and 2020, over 600 new species of plants and animals were discovered in the Amazon alone.

Activities

- 1 Suggest how the vegetation in [Figure 3.15](#) is adapted to conditions in the rainforest.
- 2 Explain how animals are adapted to living in tropical rainforests.



▲ **Figure 3.15** A simplified food chain from the tropical rainforest